

Hardik

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EDUCATION

Indian Institute of Technology Roorkee (IIT Roorkee)

Bachelor of Technology in Electrical Engineering, CGPA 8.06/10

Roorkee, Uttarakhand

July 2021 – May 2025

PROFESSIONAL SKILLS

Programming Languages: C/C++, Python, TypeScript, SQL, HTML, CSS

Frameworks & Libraries: React.js, Node.js, Express.js, Socket.IO

Databases: DBMS, MySQL, MongoDB

Tools and Technologies: Git, GitHub, VS Code, AWS (EC2, S3), GitHub Copilot, REST APIs, Docker

Technical Skills: DSA, OOPS, Operating System, Networking, Low-Level Design

WORK EXPERIENCE

Samsung R&D Institute - Bengaluru

Software Engineer

July 2025 – Present

Bengaluru, India

- Authored Design Docs for software systems, developing expertise in MAC-layer (L2) scheduling and resource allocation algorithms.
- Developed and delivered software enhancements and code changes through coding and build-test (BT) for two major releases on schedule, with a third release in progress.
- Investigated, debugged, and resolved production software issues through log analysis and root-cause identification.
- Developed an internal test automation framework, developing additional test cases to improve software test coverage.
- Executed testing activities, achieving a 5.0/5.0 test maturity score while maintaining 70%+ line coverage.

Samsung R&D Institute - Bengaluru

Software Developer Intern

June 2024 – August 2024

Bengaluru, India

- Developed Python-based parser using Regex to extract structured data from complex header files for automated Software Testing.
- Developed backend services for an internal Query Resolution Dashboard for multiple Telecom operators using Flask and MongoDB, integrating React.js for real-time updates and query resolution.
- Implemented continuous CSV-to-MongoDB data ingestion for real-time KPIs issue query rendering and resolution.

PROJECTS

RAG-Based Natural Language to MongoDB Query Generator

- Implemented a RAG pipeline for natural-language-to-MongoDB query generation, using Sentence-Transformers and Weaviate to retrieve similar question-query-schema examples.
- Integrated Gemini Pro with cross-encoder re-ranking to generate schema-aware queries, fixing errors the non-RAG baseline made (e.g., \$strLenCP for string length).
- Built a web client for entering plain-English questions and viewing the generated queries, with a RAG on/off toggle to compare outputs.

Collaborative Code Studio

- Built a real-time collaborative code editor in React.js with multi-user support, live sync, and conflict-free editing.
- Designed event-driven, WebSocket-based communication for low-latency bidirectional data exchange.
- Owned the project end-to-end — architecture through testing — validating scalability and reliability across 50+ concurrent users.

Predict US Price Closing Movements

- Predicted U.S. stock price trends using Random Forest, LSTM, and GRU on normalized time-series data.
- Tuned models and evaluated performance using MSE; LSTM-GRU achieved lowest MSE of 475.
- Demonstrated superior accuracy of deep learning for financial time-series forecasting.